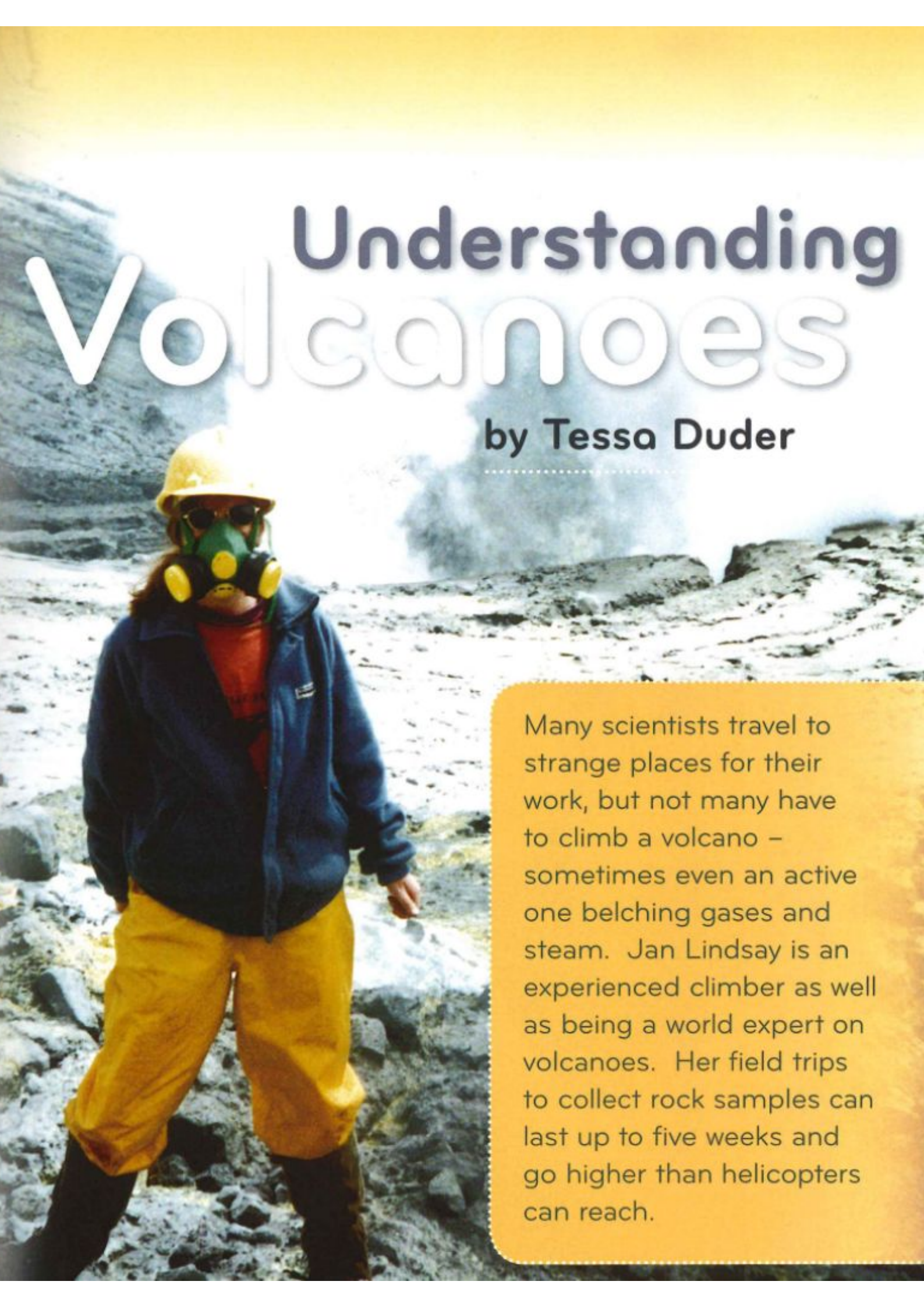




Understanding Volcanoes

by Tessa Duder

Many scientists travel to strange places for their work, but not many have to climb a volcano – sometimes even an active one belching gases and steam. Jan Lindsay is an experienced climber as well as being a world expert on volcanoes. Her field trips to collect rock samples can last up to five weeks and go higher than helicopters can reach.

PRESERVING THE SITES

When modern archaeologists are uncovering old ruins, the last thing they want to do is cause further damage. Computer scientists now work together with archaeologists at Pompeii and Herculaneum to create a database of images. These images allow the archaeologists to digitally recreate the ancient buildings, which helps them to plan an excavation. They can even model how the old Roman architects constructed the buildings.

There has been a lot of progress in finding ways to excavate ancient sites that don't destroy their treasures. The next challenge is to find better ways of protecting these cities now they have been uncovered. Sunlight and water fade murals and other artefacts. Tourists walking on the streets wear them down and sometimes they take pieces of tile or stone as souvenirs. Ivy growing in cracks makes walls and plaster crumble.

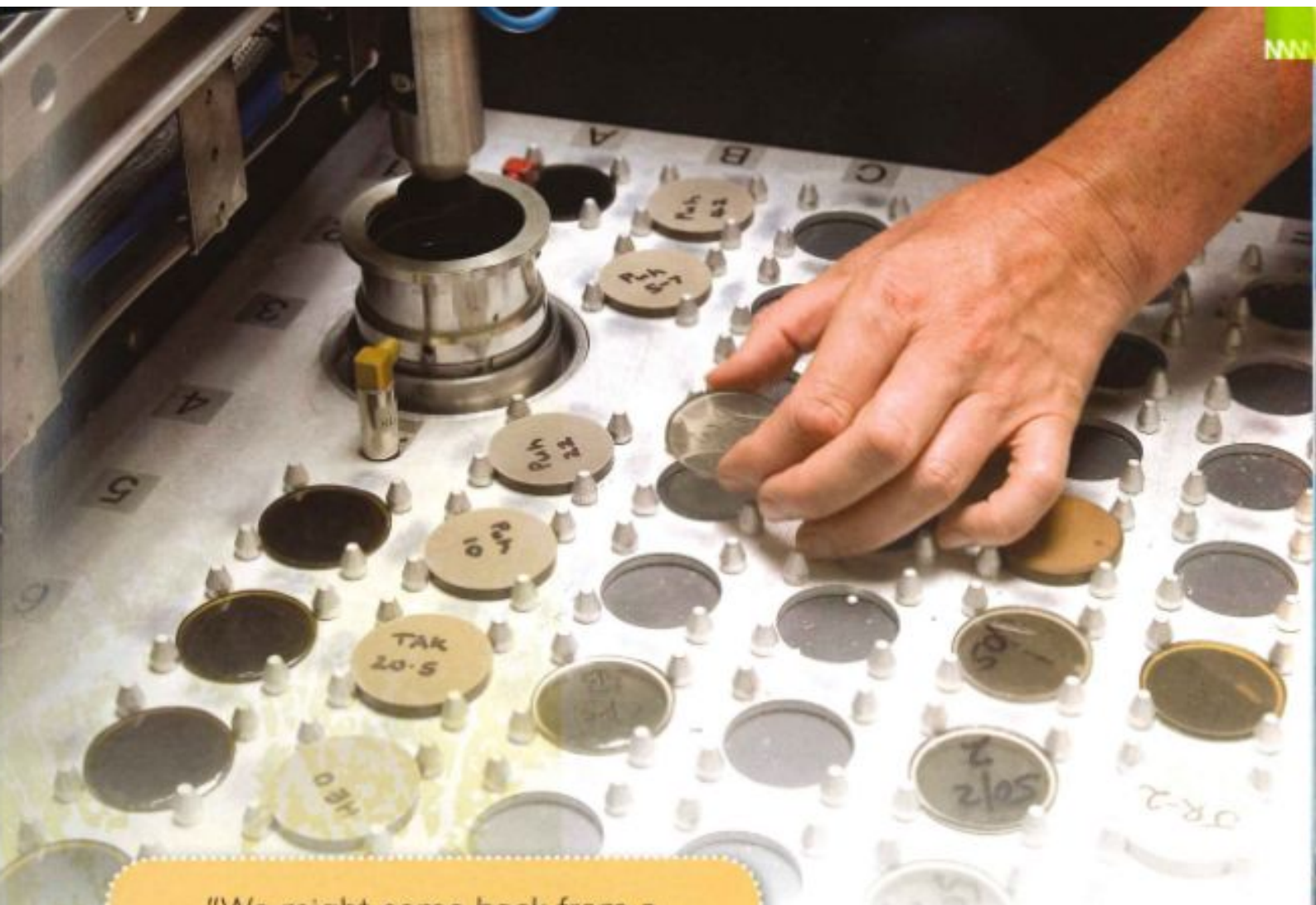
Volcanic ash protected Pompeii and Herculaneum for nearly 2000 years. Now that we have opened these amazing time capsules, we need science to help us preserve them for the people who come after us.





Jan packs a hammer, safety glasses, a hand lens, sample bags, and detailed maps for her field trips. When she gets to a site, she carefully notes where each sample is found and makes other useful observations about the site and the area.





Placing samples into
an X-ray machine

"We might come back from a field trip with 50 to 100 samples, maybe two from each site. We look for rock outcrops, where layers of lava and volcanic ash can tell us lots about past eruptions."

Back at Auckland University, the samples are crushed, melted and cooled, then zapped in an X-ray machine. This analyses the chemical elements that make up the rock. These elements provide vital clues for the detective work needed to work out a volcano's history and the possibility of future eruptions.



The highest Jan has climbed in search of rocks was 5000 metres on a volcano called La Pacana, in Chile. "That's where the outcrops were. We got the rocks, but we got altitude sickness too."

Jan's most dangerous trip was when she was helicoptered into the crater of Mount Ruapehu five months after its spectacular eruption in September 1995. "It was still rumbling, but we wanted gas samples. These can tell us if the magma is still rising up inside the mountain and if another eruption is likely. It was pretty exciting. The fumes were overwhelming, so we had to wear gas masks."

Jan first became interested in geology on a school visit to a lava cave. She went on to study in Germany, and later she worked on volcanoes in Chile and the Caribbean islands. Now she studies Auckland volcanoes like Rangitoto, Mount Eden (Maungawhau), and Brown's Island (Motukorea). She's especially interested in the impact that volcanoes can have on cities and people.



La Pacana, Chile



La Pacana, Chile



Atacama Desert, Chile



"We are always learning new things. For example, we now have proof that Rangitoto has erupted twice since Māori settled here. There will probably be another eruption somewhere in Auckland, one day, but maybe not for hundreds or even thousands of years."

Jan works closely with organisations such as Civil Defence and the Earthquake Commission to monitor the Auckland area and prepare for any volcanic event.

Jan enjoys her job because nature fascinates her. "I'm just blown away by things like, for example, the fossil forest preserved in lava and ash just off Takapuna beach. The forest was buried by the Pupuke eruption about 200 000 years ago!

That's amazing, isn't it?"

